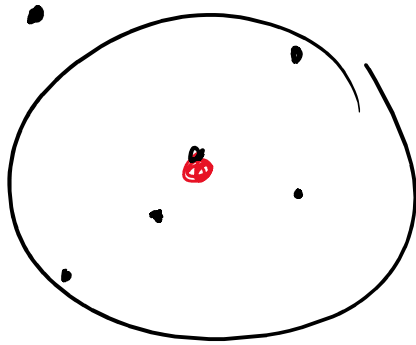
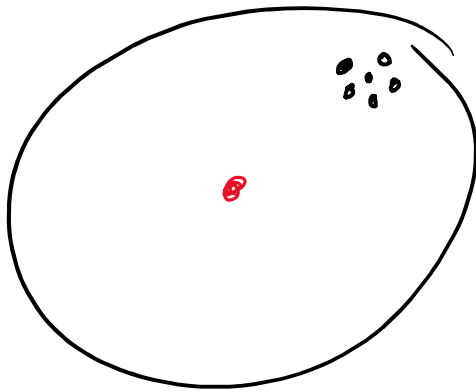
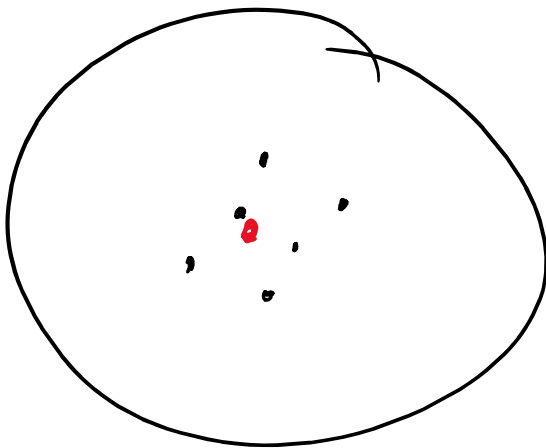




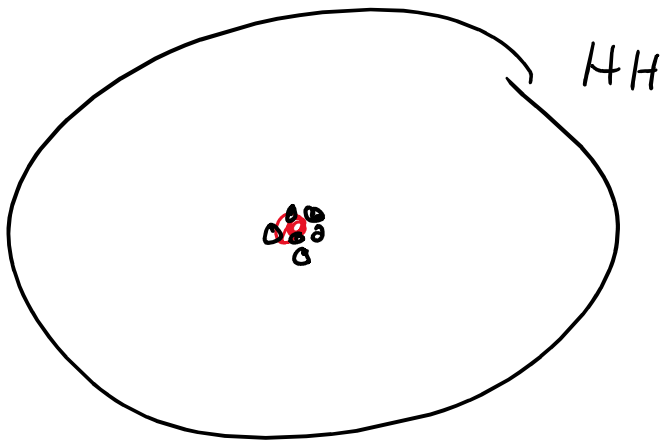
LL



LH



HL



$$P_G = \frac{1}{\sigma\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)}$$

3a.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

the max is found at $f(x=\mu)$

$$\Rightarrow f(\mu) = \frac{1}{\sigma\sqrt{2\pi}} e^{(0)}$$

$$\Rightarrow f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$$

for $f(x) = \frac{1}{2\sigma\sqrt{2\pi}}$, what is x ?

$$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) = \frac{1}{2\sigma\sqrt{2\pi}}$$

$$\Rightarrow \ln\left(\frac{1}{2}\right) = \ln\left(\exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)\right)$$

$$\Rightarrow \ln\left(\frac{1}{2}\right) = -\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2$$

$$\Rightarrow 2\sigma\ln(2) = (x-\mu)^2$$

$$\Rightarrow \pm\sqrt{2\sigma\ln(2)} = x-\mu$$

$$\Rightarrow \mu \pm \sqrt{2\sigma\ln(2)} = x$$

The FWHM = the total distance between these two values,

$$\therefore \text{FWHM} = (\mu + \sqrt{2\sigma\ln(2)}) - (\mu - \sqrt{2\sigma\ln(2)})$$

$$\Rightarrow \text{FWHM} = 2\sqrt{2\sigma\ln(2)}$$

$$2\sqrt{2\ln(2)} \approx 2.354$$

$$\Rightarrow \sqrt{2\ln(2)} \approx 1.177$$

$$2.12 \sigma \approx 2.354$$

$$\Rightarrow \boxed{\text{FWHM} \approx 2.354 \sigma}$$

3b.

$$y = b x^m$$

$$\Rightarrow \log(y) = \log(b x^m)$$

$$\Rightarrow \log(\underbrace{y}_Y) = \underbrace{m}_{\text{slope}} \underbrace{\log(x)}_X + \underbrace{\log(b)}_{\text{y-intercept } B}$$

$$\Rightarrow Y = MX + B$$